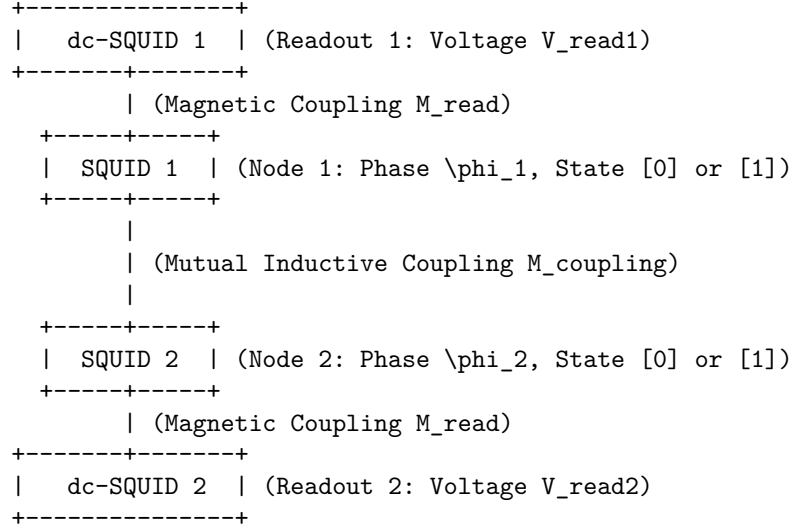


Within the flat 6D Conformal Spacetime Algebra system ($Cl(4, 1, 1)$), a 2-qubit register is physically modeled using Josephson junctions and superconducting loops (SQUIDs). In this local realist formulation, information is stored as the direction of circulating currents of phase-locked electron solitons. Operations and readouts are governed entirely by classical circuit dynamics, Maxwell's equations (with the convective D operator [2.1]), and the classical Josephson relations.

Section 13: Superconducting Josephson-Junction 2-Qubit Register

This register consists of two coupled superconducting loops interrupted by Josephson junctions (rf-SQUIDs) [Node 1 and Node 2]. The state of each node is read out using a dual-junction magnetometer (dc-SQUID) magnetically coupled to each loop.



13.1 Physical Node Design (SQUID Resonator) Each node $j \in \{1, 2\}$ is realized as a superconducting loop of inductance L interrupted by a single Josephson junction of critical current I_{c0} .

The superconducting loop carries a macroscopic current of phase-locked electron solitons [4.1]. The magnetic field bivector B inside the loop must satisfy the classical flux quantization condition, which is a topological boundary condition of the single-turn loop phase coherence:

$$\oint B \cdot I_3 dl = N\Phi_0$$

where: $* I_3 = e_{123}$ is the spatial pseudo-scalar [2.2]. $* \Phi_0 = \frac{h}{2e}$ is the magnetic

flux quantum, arising from the single-turn phase-lock constraint ($\oint \nabla \theta \cdot dl = 2\pi N$) of the circulating electron solitons [4.1] around the superconducting loop.

The phase difference φ_j across the junction is locked to the magnetic flux Φ_j threading the loop:

$$\varphi_j = 2\pi \frac{\Phi_j}{\Phi_0}$$

The potential energy $U(\varphi_j)$ of the loop under an external magnetic bias flux $\Phi_{\text{ext}} \approx \Phi_0/2$ is:

$$U(\varphi_j) = -E_J \cos \varphi_j + \frac{(\Phi_j - \Phi_{\text{ext}})^2}{2L}$$

where $E_J = \frac{I_{c0}\Phi_0}{2\pi}$ is the Josephson coupling energy. For these parameters, the potential $U(\varphi_j)$ forms a symmetric double-well. The two stable minima of this classical potential represent the logical states of the node: * **State** [0]: Clockwise circulating current of solitons ($\varphi_j \approx -\pi/2$, flux $\Phi_j \approx 0$). * **State** [1]: Counter-clockwise circulating current of solitons ($\varphi_j \approx +\pi/2$, flux $\Phi_j \approx \Phi_0$).

13.2 Josephson Junction Weak Link Dynamics The Josephson junction consists of two superconducting regions separated by a thin insulating oxide barrier of width d comparable to the close-range mechanical softening parameter [Section 3]:

$$d \sim \delta_{\text{junction}}$$

Because the barrier is narrow, the bivector field envelopes F_{total} [2.2] of the electron solitons overlap slightly across the insulator. The current of solitons crossing the junction is determined by the phase difference $\varphi_j(t) = \theta_1(t) - \theta_2(t)$ across the weak link:

$$I_j(t) = I_{c0} \sin(\varphi_j(t))$$

The voltage $V_j(t)$ developed across the barrier is driven by the rate of phase precession of the soliton fields:

$$V_j(t) = \frac{\Phi_0}{2\pi} \frac{d\varphi_j}{dt}$$

13.3 Joint Register State and Mutual Inductive Coupling The 2-qubit register consists of Node 1 and Node 2. The joint state of the register is represented as the combined magnetic flux state (Φ_1, Φ_2) of the two loops.

The coupling between the two nodes is implemented by a physical mutual inductance M_{coupling} between the two loops. The circulating current I_1 in Node 1 induces a coupling flux in the loop of Node 2:

$$\Phi_{1 \rightarrow 2} = M_{\text{coupling}} I_1$$

13.3.1 Local Realist CNOT Operation To execute a controlled-NOT (CNOT) operation, we apply a local external magnetic bias pulse to Node 1. 1. If Node 1 is in state $[0]$ (low clockwise current I_1), the induced flux $\Phi_{1 \rightarrow 2}$ is negligible. The potential barrier of Node 2 remains high, and its state is unchanged. 2. If Node 1 is in state $[1]$ (high counter-clockwise current I_1), the induced flux $\Phi_{1 \rightarrow 2}$ shifts Node 2's potential energy profile, lowering the central barrier of its double-well potential. 3. By simultaneously applying a localized RF magnetic drive pulse to Node 2 at its resonant frequency ω_{res} , the phase φ_2 precesses across the lowered barrier, driving a transition from state $[0]_2 \rightarrow [1]_2$ (or $[1]_2 \rightarrow [0]_2$) over the gate interval Δt .

This local, deterministic magnetic coupling physically executes the CNOT gate.

13.4 Non-Invasive Voltage Readout (dc-SQUID Magnetometers) To read out the state of each node, we magnetically couple each SQUID loop to an independent dc-SQUID magnetometer.

A dc-SQUID consists of a superconducting loop with two parallel Josephson junctions biased by a constant external current I_b slightly above $2I_{c0}$. The critical current of this readout loop is a periodic function of the target node's magnetic flux Φ_j :

$$I_c(\Phi_j) = 2I_{c0} \left| \cos \left(\pi \frac{\Phi_j}{\Phi_0} \right) \right|$$

Because the bias current I_b is fixed, any change in the critical current I_c (driven by the flux state of the coupled node) modulates the voltage drop $V_{\text{read},j}(t)$ across the dc-SQUID:

$$V_{\text{read},j}(t) = R_{\text{shunt}} \sqrt{I_b^2 - I_c(\Phi_j)^2}$$

By monitoring the voltage signal $V_{\text{read},j}(t)$ using standard low-noise amplifiers:

* A low voltage drop ($V_{\text{read},j} \approx 0$) corresponds to state $[0]$ (flux $\Phi_j \approx 0$). * A high voltage drop ($V_{\text{read},j} > 0$) corresponds to state $[1]$ (flux $\Phi_j \approx \Phi_0$).

This completes the model of the 2-qubit register, establishing both storage, coupling operations, and readout purely through classical, local circuit variables, Josephson precessional phases, and voltage signals.